Kernel Density Estimation (KDE) in 2D

Intuition: How KDE Works (2D)

- Imagine we have N data points in the plane:

$$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \ldots, \mathbf{x}_N, \qquad \mathbf{x}_j \in \mathbb{R}^2.$$

- Kernel Density Estimation (KDE) builds a **smooth pdf** from data.
- Instead of plotting them as spikes, we **smooth them out**.
- For each point $\mathbf{x}_j$:
  - Place a small smooth surface (kernel bump) centered at $\mathbf{x}_j$.
  - The bandwidth h_j decides how wide the bump is.
- Finally, we **add up all bumps**:
  - Where bumps overlap, the surface is tall.
  - Where there are fewer bumps, the surface is low.
- Let $\kappa(\mathbf{x})$ be the function that denotes a bump at origin $(0, 0)$.
- A suitable choice for $\kappa(.)$ is to define it as the density function of the bivariate Gaussian random variable with zero mean and covariance matrix Σ.

$$\kappa(\mathbf{x}) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2}\mathbf{x}^\top \Sigma^{-1} \mathbf{x}\right)$$

Bandwidth Parameterizations for One Kernel Bump (2D)

For a data point $\mathbf{x}_j \in \mathbb{R}^2$, a single KDE bump can be built with different levels of flexibility.

1) Isotropic bandwidth (1 unknown per bump)

$$\Sigma_j = h_j^2 I$$

Let $\kappa_j(\mathbf{x})$ denote the kernel bump function contributed by the jth data point.

$$\kappa_j(\mathbf{x}) = \frac{1}{h_j^2}\,\kappa\Big(\frac{\mathbf{x}-\mathbf{x}_j}{h_j}\Big)$$

- Same spread in all directions
- Circular level sets
- **Unknowns per bump: 1** (h_j)

Bandwidth Parameterizations for One Kernel Bump (2D) (contd..)

2) Axis-aligned bandwidth (2 unknowns per bump)

$$\Sigma_j = \begin{bmatrix} h_{j1}^2 & 0 \\ 0 & h_{j2}^2 \end{bmatrix}$$

- Different spread along x and y
- Elliptical level sets (no rotation)
- **Unknowns per bump: 2** (h_{j1}, h_{j2})

Bandwidth Parameterizations for One Kernel Bump (2D) (contd..)

3) Full covariance bandwidth (3 unknowns per bump)

$$\Sigma_j = \begin{bmatrix} \sigma_{j11} & \sigma_{j12} \\ \sigma_{j12} & \sigma_{j22} \end{bmatrix}$$

- Arbitrary orientation and spread
- Fully rotated ellipses
- **Unknowns per bump: 3**

This slide deck uses: Isotropic bandwidth (Case 1).

Kernel and Bandwidth

- Kernel $\kappa(\mathbf{u})$: symmetric pdf on $\mathbb{R}^2$ (e.g. Gaussian).
- Bandwidth $h_j > 0$: controls bump width.
  - Small h_j: narrow and spiky.
  - Large h_j: wide and flat.

$$\widehat{f}(\mathbf{x}; \mathbf{h}) = \frac{1}{N} \sum_{j=1}^{N} \frac{1}{h_j^2} \kappa\Big(\frac{\mathbf{x} - \mathbf{x}_j}{h_j}\Big).$$

From Kernel PDF to KDE CDF

- Recall the variable-bandwidth KDE (pdf):

$$\widehat{f}(\mathbf{x};\mathbf{h}) = \frac{1}{N}\sum_{j=1}^{N}\frac{1}{h_j^2}\,\kappa\Big(\frac{\mathbf{x}-\mathbf{x}_j}{h_j}\Big).$$

- The CDF at point $\mathbf{z} = (z_1, z_2)^\top$ is

$$\widehat{F}(\mathbf{z};\mathbf{h}) = \int_{-\infty}^{z_1}\int_{-\infty}^{z_2}\widehat{f}(t_1, t_2;\mathbf{h})\,dt_2\,dt_1.$$

- Substitute the KDE:

$$\widehat{F}(\mathbf{z};\mathbf{h}) = \int_{-\infty}^{z_1}\int_{-\infty}^{z_2}\frac{1}{N}\sum_{j=1}^{N}\frac{1}{h_j^2}\kappa\Big(\frac{\mathbf{t}-\mathbf{x}_j}{h_j}\Big)\,d\mathbf{t}.$$

- Swap sum and integral (finite sum — allowed).

From Kernel PDF to KDE CDF

- Swap sum and integral:

$$\widehat{F}(\mathbf{z};\mathbf{h}) = \frac{1}{N}\sum_{j=1}^{N}\int_{-\infty}^{z_1}\int_{-\infty}^{z_2}\frac{1}{h_j^2}\kappa\Big(\frac{\mathbf{t}-\mathbf{x}_j}{h_j}\Big)\,d\mathbf{t}.$$

- For each term do a change of variables:

$$\mathbf{u} = \frac{\mathbf{t}-\mathbf{x}_j}{h_j} \quad \Rightarrow \quad \mathbf{t} = \mathbf{x}_j + h_j\mathbf{u}, \qquad d\mathbf{t} = h_j^2 d\mathbf{u}.$$

- Limits transform as:

$$t_1 \le z_1 \Rightarrow u_1 \le \frac{z_1 - x_{j1}}{h_j}, \qquad t_2 \le z_2 \Rightarrow u_2 \le \frac{z_2 - x_{j2}}{h_j}.$$

- Substitute into each integral:

$$\int \frac{1}{h_j^2}\kappa(\mathbf{u})(h_j^2 d\mathbf{u}) = \int \kappa(\mathbf{u})\,d\mathbf{u}.$$

Notice the h_j^2 factors cancel.

From Kernel PDF to KDE CDF

- The remaining integral is the kernel CDF:

$$K(\mathbf{a}) = \int_{-\infty}^{a_1} \int_{-\infty}^{a_2} \kappa(\mathbf{u})\, du_2\, du_1.$$

- Final KDE-CDF formula:

$$\boxed{\widehat{F}(\mathbf{z};\mathbf{h}) = \frac{1}{N} \sum_{j=1}^{N} K\Big(\frac{\mathbf{z} - \mathbf{x}_j}{h_j}\Big)}$$

Matrix Form

- Evaluate CDF at points $\mathbf{z}_1, \ldots, \mathbf{z}_M$.
- Define the matrix:

$$\left[K_{\mathrm{mat}}(\mathbf{h})\right]_{k,j} = K\left(\frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}\right).$$

- Then

$$\widehat{\mathbf{F}}(\mathbf{h}) = \frac{1}{N} K_{\mathrm{mat}}(\mathbf{h})\, \mathbf{1}_N.$$

Matching the Empirical CDF (2D)

- Compare CDFs at evaluation points $\mathbf{z}_k$.
- Empirical CDF targets:

$$T_k = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{x_{i1} \leq z_{k1},\ x_{i2} \leq z_{k2}\}.$$

- Collect as a vector:

$$\mathbf{T} = (T_1, \ldots, T_M)^\top.$$

- Loss function:

$$J(\mathbf{h}) = \|\widehat{\mathbf{F}}(\mathbf{h}) - \mathbf{T}\|_2^2.$$

Gradient Derivation: Step by Step

- Residual:

$$\mathbf{r}(\mathbf{h}) = \widehat{\mathbf{F}}(\mathbf{h}) - \mathbf{T}.$$

- Loss:

$$J(\mathbf{h}) = \mathbf{r}(\mathbf{h})^\top \mathbf{r}(\mathbf{h}).$$

- Differentiate:

$$\frac{\partial J}{\partial h_j} = 2\,\mathbf{r}^\top \frac{\partial \widehat{\mathbf{F}}}{\partial h_j}.$$

Gradient Derivation: Step by Step

- CDF vector:

$$\widehat{\mathbf{F}}(\mathbf{h}) = \frac{1}{N} K_{\mathrm{mat}}(\mathbf{h}) \mathbf{1}_N.$$

- Differentiate:

$$\frac{\partial \widehat{\mathbf{F}}}{\partial h_j} = \frac{1}{N} \frac{\partial K_{\mathrm{mat}}}{\partial h_j} \mathbf{1}_N.$$

- Gradient:

$$\boxed{\frac{\partial J}{\partial h_j} = \frac{2}{N} \mathbf{r}^\top \left(\frac{\partial K_{\mathrm{mat}}}{\partial h_j} \mathbf{1}_N \right)}$$

Derivative of Entries (2D Kernel)

- Single matrix entry:

$$[K_{\mathrm{mat}}]_{k,j} = K\Big(\frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}\Big).$$

- Let

$$\mathbf{a}_{k,j} = \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}.$$

- Chain rule:

$$\frac{\partial}{\partial h_j} K(\mathbf{a}_{k,j}) = \nabla K(\mathbf{a}_{k,j})^\top \frac{\partial \mathbf{a}_{k,j}}{\partial h_j}.$$

- Inner derivative:

$$\frac{\partial \mathbf{a}_{k,j}}{\partial h_j} = -\frac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}.$$

Derivative of Entries (2D Kernel)

- Final simplified form:

$$\boxed{\frac{\partial}{\partial h_j} K\left(\frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}\right) = -\nabla K(\mathbf{a}_{k,j})^\top \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}}$$

Structure of $\dfrac{\partial K_{\text{mat}}}{\partial h_j}$

Definition.

$$\big(K_{\text{mat}}(\mathbf{h})\big)_{k,\ell} = K\Big(\frac{\mathbf{z}_k - \mathbf{x}_\ell}{h_\ell}\Big), \qquad K_{\text{mat}}(\mathbf{h}) \in \mathbb{R}^{M\times N}.$$

Differentiate w.r.t. h_j:

$$\frac{\partial}{\partial h_j}\big(K_{\text{mat}}(\mathbf{h})\big)_{k,\ell} = \begin{cases} 0, & \ell \neq j, \\ -\nabla K\Big(\dfrac{\mathbf{z}_k - \mathbf{x}_j}{h_j}\Big)^\top \dfrac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}, & \ell = j. \end{cases}$$

Matrix structure. Only the j-th column is nonzero.

Structure of $\dfrac{\partial K_{\text{mat}}}{\partial h_j}$

$$\frac{\partial K_{\text{mat}}}{\partial h_j} = \begin{bmatrix} 0 & \cdots & -\nabla K(\mathbf{a}_{1,j})^\top \dfrac{\mathbf{z}_1 - \mathbf{x}_j}{h_j^2} & \cdots & 0 \\ 0 & \cdots & -\nabla K(\mathbf{a}_{2,j})^\top \dfrac{\mathbf{z}_2 - \mathbf{x}_j}{h_j^2} & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & -\nabla K(\mathbf{a}_{M,j})^\top \dfrac{\mathbf{z}_M - \mathbf{x}_j}{h_j^2} & \cdots & 0 \end{bmatrix}$$

where

$$\mathbf{a}_{k,j} = \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}.$$

Substitute Entrywise Derivative into the Gradient

Recall:

$$\frac{\partial J}{\partial h_j} = \frac{2}{N}\,\mathbf{r}(\mathbf{h})^\top \Big(\frac{\partial K_{\mathrm{mat}}(\mathbf{h})}{\partial h_j}\mathbf{1}_N\Big), \qquad \mathbf{r}(\mathbf{h}) = \widehat{\mathbf{F}}(\mathbf{h}) - \mathbf{T}.$$

Form the vector $\dfrac{\partial K_{\mathrm{mat}}}{\partial h_j}\mathbf{1}_N$:

$$\frac{\partial K_{\mathrm{mat}}}{\partial h_j}\mathbf{1}_N = \begin{bmatrix} -\nabla K(\mathbf{a}_{1,j})^\top \dfrac{\mathbf{z}_1 - \mathbf{x}_j}{h_j^2} \\ -\nabla K(\mathbf{a}_{2,j})^\top \dfrac{\mathbf{z}_2 - \mathbf{x}_j}{h_j^2} \\ \vdots \\ -\nabla K(\mathbf{a}_{M,j})^\top \dfrac{\mathbf{z}_M - \mathbf{x}_j}{h_j^2} \end{bmatrix} \in \mathbb{R}^M.$$

Substitute Entrywise Derivative into the Gradient

Inner product with residual:

$$\mathbf{r}(\mathbf{h})^\top \left(\frac{\partial K_{\mathrm{mat}}}{\partial h_j} \mathbf{1}_N \right) = -\sum_{k=1}^{M} r_k(\mathbf{h}) \, \nabla K(\mathbf{a}_{k,j})^\top \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}.$$

Component form:

$$\boxed{\frac{\partial J}{\partial h_j} = -\frac{2}{N} \sum_{k=1}^{M} r_k(\mathbf{h}) \, \nabla K(\mathbf{a}_{k,j})^\top \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}}$$

Matrix-Vector Compact Form

Define the $M \times N$ matrix $D(\mathbf{h})$ with entries

$$D_{k,j}(\mathbf{h}) = -\nabla K\Big(\frac{\mathbf{z}_k - \mathbf{x}_j}{h_j}\Big)^\top \frac{\mathbf{z}_k - \mathbf{x}_j}{h_j^2}.$$

Then the gradient becomes

$$\boxed{\nabla_{\mathbf{h}} J(\mathbf{h}) = \frac{2}{N}\, D(\mathbf{h})^\top \mathbf{r}(\mathbf{h})}$$

Gradient Descent Update

- Update rule:

$$\mathbf{h}^{(t+1)} = \mathbf{h}^{(t)} - \eta \, \nabla_{\mathbf{h}} J(\mathbf{h}^{(t)}).$$

- To keep $h_j > 0$, use log-variables:

$$\alpha_j = \log h_j, \qquad \alpha_j^{(t+1)} = \alpha_j^{(t)} - \eta \, \frac{\partial J}{\partial \alpha_j}.$$

Computations per Gradient Step (2D KDE)

Gradient formula:

$$\nabla_{\mathbf{h}} J(\mathbf{h}) = \frac{2}{N} D(\mathbf{h})^\top \mathbf{r}(\mathbf{h})$$

Each gradient step requires the following computations.

1: $K_{mat}1$

$$\widehat{\mathbf{F}} = \frac{1}{N} K_{\mathrm{mat}} \mathbf{1}_N$$

- Matrix–vector multiplication
- Produces density estimates at M points

Cost: $\mathcal{O}(MN)$

2: $D^T r$

$$\nabla_{\mathbf{h}} J = \frac{2}{N} D^\top \mathbf{r}$$

- Matrix–vector multiplication
- Produces gradient for all h_j

Cost: $\mathcal{O}(MN)$

Use HODLR matrix-vector product to speed up both matrix-vector products